

BRIEF REPORT

Uniparental disomy causes deficiencies of vitamin K-dependent proteins

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Essentials

- Vitamin K-dependent coagulant factor deficiency (VKCFD) is a rare autosomal recessive disorder.
- We describe a case of inherited VKCFD due to uniparental disomy.
- The homozygous mutation caused the absence of GGCX isoform 1 and overexpression of $\Delta 2$ GGCX.
- Hepatic and non-hepatic vitamin K-dependent proteins must be assayed to monitor VKCFD treatment.

Summary. *Background:* Inherited deficiency of all vitamin K-dependent coagulant factors (VKCFD) is a rare autosomal recessive disorder caused by mutations in the γ -glutamyl carboxylase gene (*GGCX*) or the vitamin K epoxide reductase gene (*VKORC1*), with great heterogeneity in terms of both clinical presentation and response to treatment. *Objective:* To characterize the molecular basis of VKCFD in a Spanish family. *Methods and Results:* Sequencing of candidate genes, comparative genomic hybridization and massive sequencing identified a new mechanism causing VKCFD in the proband. Uniparental disomy (UPD) of chromosome 2 caused homozygosity of a mutation (c.44-1G>A) resulting in aberrant *GGCX* splicing. This change contributed to absent expression of the mRNA coding for the full-length protein, and to four-fold overexpression of the smaller

mRNA isoform lacking exon 2 ($\Delta 2$ GGCX). $\Delta 2$ GGCX might be responsible for two unexpected clinical observations in the patient: (i) increased plasma osteocalcin levels following vitamin K₁ supplementation; and (ii) a mild non-bleeding phenotype. *Conclusions:* Our study identifies a new autosomal disease, VKCFD1, caused by UPD. These data suggest that the $\Delta 2$ GGCX isoform may retain enzymatic activity, and strongly encourage the evaluation of both hepatic and non-hepatic vitamin K-dependent proteins to assess differing responses to vitamin K supplementation in VKCFD patients.

Keywords: alternative splicing; γ -glutamyl carboxylase; uniparental disomy; vitamin K; VKCFD1.

Introduction

Deficiency of all vitamin K (VK)-dependent (VKD) coagulant factors (VKCFD) is a rare, autosomal recessive inherited disorder caused by mutations in the γ -glutamyl carboxylase gene (*GGCX*) or the VK epoxide reductase gene (*VKORC1*) [1,2]. VKCFDs have particular relevance to the hemostatic system, as both procoagulant (factor II, FVII, FIX, and FX) and anticoagulant (protein C, protein S, and protein Z) factors are VKD proteins [3]. Proteins involved in bone and heart development, such as matrix Gla protein, and osteocalcin, are also VKD proteins [4,5]. Finally, Gas6, PRGP1 and PRGP2 also belong to the VKD protein family, but have physiologic functions that are as yet not fully characterized [6,7]. Accordingly, multiple disorders, including abnormal bone and heart development, and coagulopathies, may be expected in VKCFD patients [8,9]. Intriguingly, VKCFD patients show heterogeneous clinical phenotypes with respect to the number of affected systems and their clinical severity [1,8,9]. Additionally,

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the effectiveness of supplementation with VK is variable [1,8,9].

Methods

A Caucasian infant suspected of having VKCFD and his relatives were studied. Informed consent was obtained from all participants.

Hemostatic VKD proteins and coagulation parameters were determined in an automated coagulometer.

GLA-type osteocalcin (GLA-OC) and undercarboxylated osteocalcin (UC-OC) were quantified by ELISA (Takara, Condalab, Madrid, Spain).

Experienced pediatricians performed dysmorphicologic, bone mass density and radiologic examinations throughout the patient's growth.

The promoter region and the coding and flanking regions of both *VKORC1* (NG_011564.1) and *GGCX* (NG_011811.2) were sequenced from genomic DNA, as described previously [10,11]. RT-PCR on whole RNA obtained from mononuclear cells of peripheral blood of healthy subjects, the patient and family members was used for sequencing and quantification of *GGCX* isoforms. This study was also performed with mRNA obtained from the livers of three healthy subjects (Research Center of Experimental Pathology Department of La Fe Hospital and CIBERehd, Valencia, Spain). Quantitative RT-qPCR, for which *ACTB* was used as an endogenous reference, was performed with SYBR Green master mix (Life Technologies, Madrid, Spain). Quantification was performed with the $2^{-\Delta C_t}$ method. The primers used for molecular analysis of *GGCX* are shown in Fig. 1A.

DNA was also analyzed on the Affymetrix CytoScan 750K Array according to the manufacturer's protocol (Affymetrix, Santa Clara, CA, USA). The CytoScan 750K Array provides whole-genome coverage, with > 750 000 markers for copy number analysis and > 200 000 single-nucleotide polymorphisms (SNPs) for genotyping. Data analysis was performed with the CHROMOSOME ANALYSIS SUITE (CHAS) (Affymetrix). All markers were used to determine copy number, and SNP probes were used to calculate loss of heterozygosity (LOH) and genotype calls. In addition, an Illumina TruSight One sequencing panel on a MiSeq platform (Illumina, San Diego, CA, USA) was used. This kit enables enrichment and final analysis of a panel of ~ 5000 genes (<http://www.illumina.com/products/trusight-one-sequencing-panel.html>). The Trusight One sequencing panel contains all of the reagents necessary for the amplification, enrichment and indexing of the samples. Each procedure was performed according to the manufacturer's instructions. The Qubit 2.0 Fluorometer system (Life Technologies) was used for the quantification of the DNA and the genomic library. For data analysis, we used VARIANTSTUDIO version 2.2.1 (Illumina) and INTEGRATIVE GENOMICS VIEWER version 2.3.68 (Broad Institute, Cambridge, MA, USA).

Results and discussion

A 14-month-old male patient was admitted for evaluation of a deltoid hematoma post-vaccination. Anamnesis excluded familial bleeding disorders and drug intake during pregnancy. The parents reported no consanguinity. Neonatal nephrocalcinosis was reported. Physical and radiologic examination excluded skeletal abnormalities, dysmorphism or ossifications at diagnosis. It is of note that the patient did not develop bleeding episodes as a preterm and low-weight infant (32 + 5 gestational weeks, 1450 g) receiving mechanical ventilation because of neonatal respiratory distress syndrome (hyaline membrane disease). Moreover, the patient had an interventricular septal defect with mild aortic insufficiency and a Wolff-Parkinson-White syndrome.

Laboratory evaluation showed severely deficient levels (< 5%) of all VKD coagulation factors (Table 1), and a high UC-OC/GLA-OC ratio at diagnosis. In contrast, non-VKD coagulation factors showed normal values (Table 1). As renal function and liver function were normal, malabsorption, liver disease, cystic fibrosis and coumarin intoxication were excluded. A diagnosis of VKCFD was established.

Treatment with VK₁ (Konakion) was initiated with different schedules (5–10 mg daily; oral or intravenous) up until the present (age 4 years; 5 mg daily; oral), but neither coagulation test findings, VKD coagulation factors nor the International Normalized Ratio (INR) (always > 8) were normalized (Table 1). However, the UC-OC/GLA-OC ratio was reduced after VK₁ treatment, although it has never reached normal values (Table 1). Serum VK levels of 300 nmol L⁻¹ were found when the patient was taking VK₁ at 15 mg daily.

During follow-up, frequent bruising and subcutaneous hematomas were reported after exposure to traumatic agents, and self-limiting episodes of bleeding in the mouth. Two episodes of oral mucosa bleeding after trauma were treated with prothrombin complex. The patient has no hemarthrosis. When the patient was aged 3 years, clinodactyly on the right hand was identified, and a new evaluation revealed calcification of the trachea and both bronchi.

No significant mutation was found in *VKORC1*. However, the proband carried a new homozygous mutation in the last nucleotide of intron 1 (c.44-1G>A) of *GGCX* that severely disturbed the acceptor sequence of intron 1 according to HUMAN SPLICING FINDER version 2.4.1-UMD software (Fig. 1C) [12]. The heterozygous mutation was present in the father and paternal grandmother (Fig. 1B, C). Unexpectedly, the mother did not carry this mutation. Interestingly, all *GGCX* polymorphisms sequenced in the proband were homozygous. The analysis of these SNPs the patient's parents revealed findings compatible with LOH in the proband (Fig. 1B,C). The comparative genome hybridization (CGH) analysis confirmed LOH for all

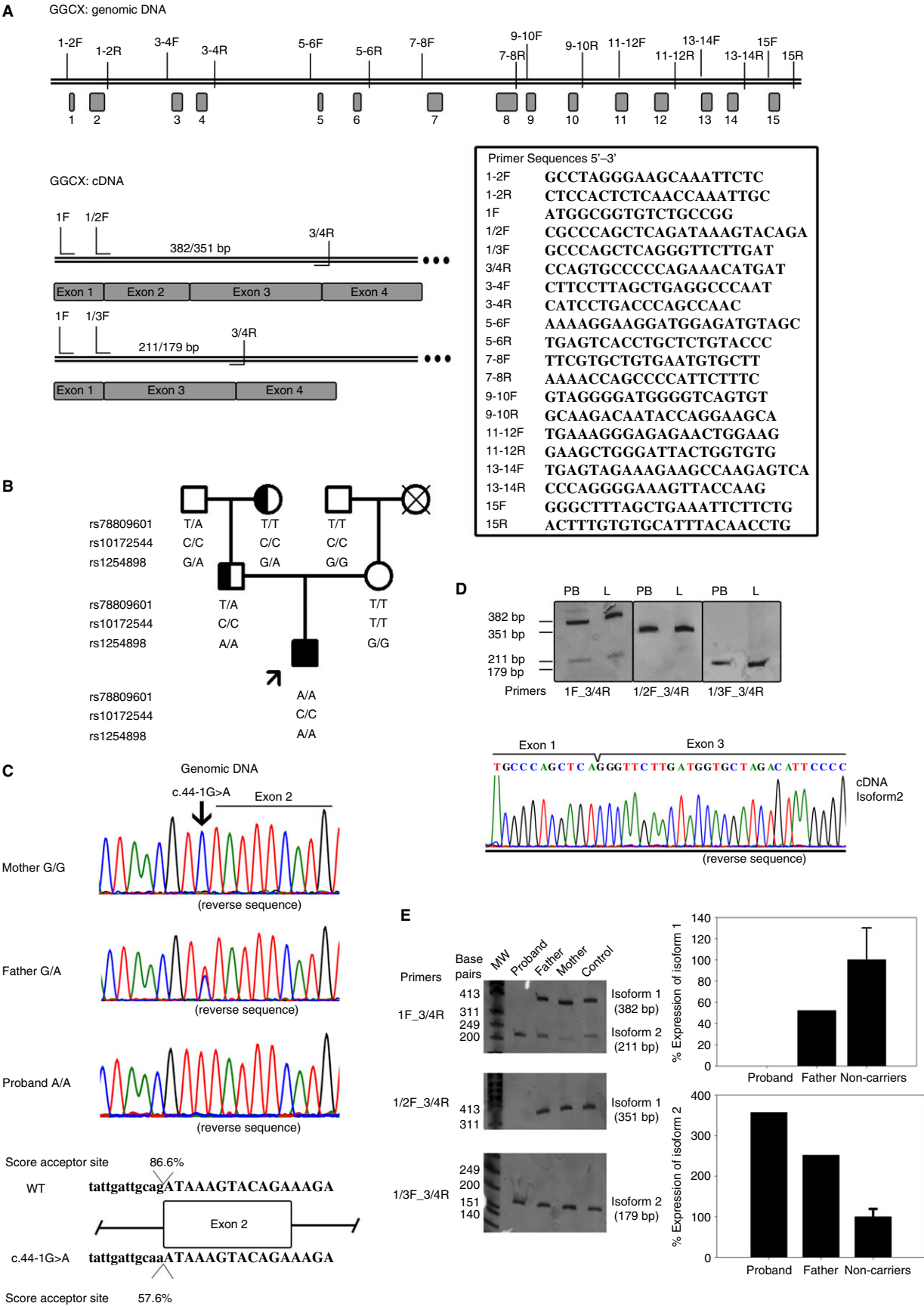


Fig. 1. Characterization of the family with VKCFD1. (A) Schematic representation of *GGCX*, showing exons and primers used for PCR amplification of *GGCX*, from genomic DNA or cDNA. (B) Family tree with genetic information on *GGCX*. The proband, who is homozygous for the c.44-1G>A mutation, is indicated by an arrow. Semi-filled squares heterozygous c.44-1G>A status. Single-nucleotide polymorphisms informative for loss of heterozygosity in *GGCX* are also indicated: rs78809601 (c.43+33T>A), rs10172544 (c. 44-162G>T), and rs1254898 (c. 44-32G>A). (C) Electropherogram of the intron 1–exon 2 boundary in the proband and his parents (reverse sequence). The prediction of the effect of this mutation on the splicing acceptor signal determined with HUMAN SPLICING FINDER is also shown. (D) Expression of mRNA of *GGCX* in liver (L) and peripheral blood (PB) of healthy subjects. The sequence of the small isoform is also shown, demonstrating the absence of exon 2 ($\Delta 2GGCX$). (E) Isoforms of *GGCX* expressed in PB of the proband and relatives. For the expression levels of the two *GGCX* mRNA isoforms, data were analyzed with the comparative threshold cycle method ($2^{-\Delta C_t}$ method), with β -actin as an endogenous reference control for quantification and normalization. MW, molecular weight marker; VKCFD, inherited deficiency of all vitamin K-dependent coagulant factors; WT, wild type. [Color figure can be viewed at wileyonlinelibrary.com.]

of chromosome 2. Table 2 shows gross defects identified by CGH. Finally, next-generation sequencing (NGS) analysis verified uniparental disomy (UPD). The proband had two copies of the paternal chromosome 2 (Table 3).

RT-PCR of *GGCX* from livers and peripheral blood (as this gene is also expressed in other tissues [13]) showed the presence of two *GGCX* isoforms in healthy subjects (Fig. 1D). Sequencing of the smallest isoform confirmed that exon 2 was absent ($\Delta 2GGCX$) (Fig. 1D). Analysis of these isoforms in peripheral blood of the

proband and relatives revealed that, in comparison with non-carriers, expression of isoform 1 was reduced two-fold in the father and absent in the proband (Fig. 1E), whereas expression of isoform 2 ($\Delta 2GGCX$) was increased almost four-fold in the proband and two-fold in the father (Fig. 1E).

The VKCFD diagnosed in this study was caused by an exceptional molecular mechanism. A UPD makes a single *GGCX* mutation pathogenic. There is increasing evidence supporting the contention that UPD underlies different

Table 1 Analytic values observed in the proband and his relatives

	Propositus		Relatives			
	Diagnosis	Vitamin K*	Father	Mother	Paternal grandfather†	Paternal grandmother
PT (s)	98.9	95.4 ± 11.9	15.4	16.1	23.1	14.9
INR	9	8.9 ± 0.8	1	1	2	ND
QI (%)	9	9.1 ± 1.3	100	93	ND	ND
APTT (s)	53.1	40.0 ± 7.3	31.4	26.1	27.0	28.5
TT (s)	15.4	67.0 ± 11.0	12.6	13.1	ND	ND
Fibrinogen (mg dL ⁻¹)	352	457.9 ± 84.5	292	395	402	391
FII (%)	2	3.7 ± 0.9	97	94	24	100
FVII (%)	1.7	3.2 ± 1.7	126	73.8	53	146
FIX (%)	4.7	9.1 ± 2.8	121	134.9	41	83
FX (%)	2	4.7 ± 1.4	98	97.8	20	95
PC (%)	3	4.1 ± 2.1	134	147	29	116
PS (%)	3.6	5.6 ± 0.8	82	75	50	100
FV (%)	164	147.3 ± 24.4	ND	ND	ND	ND
FVIII (%)	133	141.8 ± 8.5	ND	ND	ND	ND
Platelets (× 10 ⁹ L ⁻¹)	288	243	ND	ND	ND	ND
UC-OC (ng mL ⁻¹)	91.2	9.5	9.0	3.4	2.1	4.0
GLA-OC (ng mL ⁻¹)	1.1	1.0	3.7	4.5	1.3	4.7
OCr (AU)	84.5	9.5	2.4	0.8	1.6	0.9
AST (U L ⁻¹)	68	ND	ND	ND	ND	ND
ALT (U L ⁻¹)	35	ND	ND	ND	ND	ND
GGT (U L ⁻¹)	15	ND	ND	ND	ND	ND
Urea (mg dL ⁻¹)	45	ND	ND	ND	ND	ND
Creatinine (mg dL ⁻¹)	0.23	ND	ND	ND	ND	ND
VD (ng mL ⁻¹)	26.4	ND	ND	ND	ND	ND
Calcium (mg dL ⁻¹)	10.1	ND	ND	ND	ND	ND
Phosphorus (mg dL ⁻¹)	5.3	ND	ND	ND	ND	ND
<i>GGCX</i> mutation	+/+		+/-	-/-	-/-	+/-

ALT, alanine transaminase; APTT, activated partial thromboplastin time; AST, aspartate transaminase; AU, arbitrary units; GGT, γ -glutamyl transferase; GLA-OC, GLA-type osteocalcin; INR, International Normalized Ratio; ND, Not determined; OCr, osteocalcin ratio (UC-OC/GLA-OC); PC, protein C, activity; PS, protein S, antigen; PT, prothrombin time; QI, Quick Index; TT, thrombin time; UC-OC, undercarboxylated osteocalcin; VD, vitamin D. *Mean ± standard deviation of determinations throughout 3 years of follow-up. †Receiving anticoagulant therapy with acenocoumarol.

Table 2 Gross chromosome alterations identified in the proband

Type	Chromosome	Cytoband start	Cytoband end	Size (kbp)	Microarray nomenclature
LOH	1	p33	p32.3	3930.029	arr[hg19] 1p33p32.3(49,149,494-53,079,523) hmz
LOH	2	p25.3	p11.2	87 002.339	arr[hg19] 2p25.3p11.2(50,813-87,053,152) hmz
Loss	2	p16.3	p16.3	12.704	arr[hg19] 2p16.3(50,602,678-50,615,382)x1
LOH	2	p11.2	p11.2	804.137	arr[hg19] 2p11.2(88,312,761-89,116,898) hmz
Loss	2	q31.2	q31.2	66.659	arr[hg19] 2q31.2(178,812,739-178,879,398)x1
Gain	2	q11.2	q11.2	163.038	arr[hg19] 2q11.2(97,718,323-97,881,361)x4
LOH	2	q11.1	q37.3	147 222.626	arr[hg19] 2q11.1q37.3(95,550,957-242,773,583) hmz
Loss	3	p24.1	p24.1	23.552	arr[hg19] 3p24.1(27,615,814-27,639,366)x1
Loss	4	q31.22	q31.22	30.417	arr[hg19] 4q31.22(148,366,789-148,397,206)x1
Loss	6	p25.3	p25.3	77.546	arr[hg19] 6p25.3(254,253-331,799)x1
LOH	6	p22.2	p22.1	4004.683	arr[hg19] 6p22.2p22.1(25,410,953-29,415,636) hmz
Gain	7	p22.1	p22.1	42.088	arr[hg19] 7p22.1(5,948,848-5,990,936)x3
Loss	9	q21.2	q21.2	33.842	arr[hg19] 9q21.2(80,056,837-80,090,679)x1
Loss	11	p15.4	p15.4	24.653	arr[hg19] 11p15.4(5,783,740-5,808,393)x1
Loss	12	q21.32	q21.32	4.354	arr[hg19] 12q21.32(88,497,575-88,501,929)x1
Gain	12	p13.31	p13.31	58.282	arr[hg19] 12p13.31(8,536,599-8,594,881)x4
Loss	14	q32.33	q32.33	25.283	arr[hg19] 14q32.33(106,530,533-106,555,816)x0
Loss	14	q24.3	q24.3	40.687	arr[hg19] 14q24.3(73,989,216-74,029,903)x1
Loss	14	q22.1	q22.1	17.14	arr[hg19] 14q22.1(53,703,059-53,720,199)x1
Loss	14	q23.1	q23.1	3.76	arr[hg19] 14q23.1(61,158,999-61,162,759)x1
Gain	14	q32.33	q32.33	431.162	arr[hg19] 14q32.33(106,099,298-106,530,460)x3
Gain	14	q32.33	q32.33	215.998	arr[hg19] 14q32.33(106,560,962-106,776,960)x3
Gain	16	p11.2	p11.2	965.643	arr[hg19] 16p11.2(32,848,682-33,814,325)x3
Loss	17	q21.31	q21.31	62.112	arr[hg19] 17q21.31(44,135,984-44,198,096)x1
Gain	17	q21.31	q21.31	66.291	arr[hg19] 17q21.31(44,201,343-44,267,634)x4
Gain	19	q13.41	q13.41	15.531	arr[hg19] 19q13.41(53,413,117-53,428,648)x4
Loss	X	q28	q28	10.17	arr[hg19] Xq28(149,840,885-149,851,055)x0
Loss	X	p22.33	p22.33	26.568	arr[hg19] Xp22.33(3,070,689-3,097,257)x0
Loss	X	p11.21	p11.21	40.168	arr[hg19] Xp11.21(55,674,153-55,714,321)x0
Loss	X	p22.33	p22.33	0.001	arr[hg19] Xp22.33(2,696,761-2,696)x1
Loss	X	q28	q28	0.001	arr[hg19] Xq28(154,941,868-154,941,869)x1
LOH	X	p22.33	p11.1	55 482.556	arr[hg19] Xp22.33p11.1(2,744,764-58,227,320) hmz
LOH	X	q11.1	q28	93 197.176	arr[hg19] Xq11.1q28(62,036,670-155,233,846) hmz

LOH, loss of heterozygosity.

autosomal diseases, although, to date, only 40 examples of such recessive trait transmission have been reported [14]. The fact that the pathogenic mutation unmasked by the UPD affects splicing supports the relevance that mutations affecting splicing may cause multiple diseases [15].

Our data support the hypothesis that the c.44-1G>A mutation in intron 1 of *GGCX*, at least in peripheral blood, causes the loss of expression of isoform 1, which encodes the full-length protein, but induces the overexpression of isoform 2, a small mRNA without exon 2 ($\Delta 2$ GGCX). There are few data on the function of the protein encoded by $\Delta 2$ GGCX. Only one report evaluating several hepatocellular carcinoma cell lines has described $\Delta 2$ GGCX as a specific mRNA for des- γ -carboxy-prothrombin-positive cell lines [16]. This study also suggests that $\Delta 2$ GGCX could function as a dominant negative inhibitor of wild-type GGCX [16]. However, the expression of $\Delta 2$ GGCX mRNAs in livers of healthy subjects and the γ -carboxy-osteocalcin detected in the patient who only expresses $\Delta 2$ GGCX in peripheral blood argue against the suggested dominant negative effect on

wild-type GGCX. Rather, these results support the contention that $\Delta 2$ GGCX retains enzymatic activity, at least in non-hepatic tissues, which could rescue the absence of activity of full-length GGCX in the proband and potentially during anticoagulant therapy. A tissue-dependent γ -carboxylation effect has been previously suggested to be manifested by VKORC1L1, an enzyme that rescues VK-2,3-epoxide reductase activity in some extrahepatic tissues during anticoagulation therapy [17]. Interestingly, a recent study found a new VKCFD1 case with severe bleeding carrying two new mutations affecting exons 4 and 8 of *GGCX* that caused decreases in the levels of hepatic clotting factors; these were restored by treatment with VK₁, but the levels of matrix γ -carboxy-glutamate protein, a non-hepatic VKD protein, were not ameliorated by treatment with VK₁ [18]. These and our data suggest that the effect of supplementation with VK₁ largely depends on the target tissue and the *GGCX* mutation, but also on the target protein. Therefore, further studies are required to provide additional information on the function of GGCX in different tissues, and to clarify the function of $\Delta 2$ GGCX. Independently of this controversy, all of these

Table 3 Genetic defects identified by next-generation sequencing in the father, mother, and proband, located in genes of chromosome 2

Gene	Chromosome	Coordinate	HGVS	dbSNP ID	Father	Mother	Proband
<i>SNTG2</i>	2	1 320 143	NM_018968.3:c.1484C>T	rs4927635	39.29	100	0
<i>MYT1L</i>	2	1 842 968	NM_015025.2:c.3027A>G	rs6728368	54.11	0	98.49
<i>ROCK2</i>	2	11 359 120	NM_004850.3:c.1292C>A	rs2230774	0	100	0
<i>LPIN1</i>	2	11 911 761	NM_001261428.1:c.699C>T	rs11538448	0	100	0
<i>MATN3</i>	2	20 202 929	NM_002381.4:c.909G>A	rs28452699	38.6	0	100
<i>APOB</i>	2	21 225 753	NM_000384.2:c.12541G>A	rs1042031	50	0	98.65
<i>APOB</i>	2	21 232 195	NM_000384.2:c.7545C>T	rs693	51.03	100	0
<i>APOB</i>	2	21 232 804	NM_000384.2:c.6936C>T	rs1041968	45.41	100	0
<i>APOB</i>	2	21 250 914	NM_000384.2:c.1853C>T	rs679899	44.83	0	99.4
<i>APOB</i>	2	21 263 900	NM_000384.2:c.293C>T	rs1367117	62.26	100	0
<i>TP53I3</i>	2	24 300 563	NM_004881.4:c.885A>G	rs10209238	48.72	0	98.79
<i>NCOA1</i>	2	24 905 927	NM_003743.4:c.462G>C	rs11125744	46.22	100	0
<i>NCOA1</i>	2	24 974 945	NM_003743.4:c.3801T>G	rs11125763	47.54	96.55	0
<i>ADCY3</i>	2	25 141 538	NM_004036.3:c.319T>C	rs11676272	42.74	0	98.5
<i>HADHA</i>	2	26 455 127	NM_000182.4:c.474C>T	rs11552518	53.7	0	99.04
<i>HADHB</i>	2	26 502 875	NM_000183.2:c.825T>C	rs1056389	59.4	0	99.24
<i>OTOF</i>	2	26 699 126	NM_194248.2:c.2736G>C	rs4335905	99.01	0	100
<i>OTOF</i>	2	26 741 961	NM_194248.2:c.244C>T	rs13031859	42.55	100	0
<i>ALK</i>	2	29 940 529	NM_004304.4:c.702T>A	rs2246745	100	0	98.33
<i>LTBP1</i>	2	33 540 248	NM_006943.2:c.3642G>A	rs1058840	66	100	0
<i>LTBP1</i>	2	33 623 463	NM_006943.2:c.5017A>T		45.81	0	100
<i>SOS1</i>	2	39 262 348	NM_005633.3:c.1074+5G>C	rs145155424	67.27	0	100
<i>THADA</i>	2	43 519 977	NM_001083953.1:c.4814G>A	rs35720761	59.17	0	100
<i>THADA</i>	2	43 732 823	NM_001083953.1:c.3559A>G	rs7578597	41.41	0	100
<i>THADA</i>	2	43 801 712	NM_001083953.1:c.1492T>C	rs11899823	47.24	0	98.88
<i>THADA</i>	2	43 804 229	NM_001083953.1:c.969A>G	rs13021894	41.73	0	98.4
<i>THADA</i>	2	43 805 739	NM_001083953.1:c.729G>A	rs61754256	60.38	0	100
<i>THADA</i>	2	43 819 139	NM_001083953.1:c.123T>C	rs17031095	42.47	0	98.85
<i>LRPPRC</i>	2	44 177 701	NM_133259.3:c.1677+11C>G	rs58811869	0	100	0
<i>LRPPRC</i>	2	44 201 376	NM_133259.3:c.1068A>G	rs4953042	44.19	0	100
<i>LRPPRC</i>	2	44 209 477	NM_133259.3:c.246G>A	rs6741066	0	100	0
<i>EPCAM</i>	2	47 601 106	NM_002354.2:c.344T>C	rs1126497	0	100	0
<i>STON1-GTF2A1L, LHCGR</i>	2	48 915 871	NM_000233.3:c.1065T>C	rs11125179	46.98	100	0
<i>STON1-GTF2A1L, LHCGR</i>	2	48 921 375	NM_000233.3:c.935A>G	rs2293275	43.59	100	0
<i>FAM161A</i>	2	62 052 380	NM_001201543.1:c.*1210T>C	rs3736598	47.71	98.84	0
<i>FAM161A</i>	2	62 053 290	NM_001201543.1:c.*300T>C	rs6748320	50.24	100	0
<i>FAM161A</i>	2	62 053 546	NM_001201543.1:c.*41_*43delAGC	rs72380346	0	97.3	0
<i>FAM161A</i>	2	62 069 358	NM_001201543.1:c.321A>G	rs11125895	48.55	0	100
<i>CEP68</i>	2	65 296 798	NM_015147.2:c.220G>A	rs7572857	52.26	0	100
<i>CEP68</i>	2	65 298 839	NM_015147.2:c.609A>G	rs17849707	46.64	0	99.15
<i>NFU1</i>	2	69 659 126	NM_001002755.2:c.74T>A	rs4453725	0	98.55	0
<i>ASPRV1</i>	2	70 188 676	NM_152792.2:c.145A>G	rs3796097	41.3	0	99.39
<i>ATP6V1B1</i>	2	71 163 086	NM_001692.3:c.2T>C	rs11681642	100	0	100
<i>DYSF</i>	2	71 838 597	NM_001130987.1:c.4062C>A	rs2303606	63.04	100	0
<i>CYP26B1</i>	2	72 361 960	NM_019885.2:c.791T>C	rs2241057	49.3	0	100
<i>MOGS</i>	2	74 690 039	NM_006302.2:c.877C>T	rs2268416	0	99.01	0
<i>MOGS</i>	2	74 690 378	NM_006302.2:c.715G>A	rs1063588	0	96.61	0
<i>HK2</i>	2	75 113 789	NM_000189.4:c.2208C>G	rs2229627	0	100	0
<i>GGCX</i>	2	85 788 109	NM_000821.5:c.44-1G>A		53.57	0	100
<i>IMMT</i>	2	86 400 824	NM_006839.2:c.370C>T	rs1050301	43.75	0	100
<i>REEP1</i>	2	86 481 835	NM_001164730.1:c.306G>A	rs2276625	100	0	100
<i>CD8A</i>	2	87 012 073	NM_001145873.1:c.*970T>C	rs1051386	99.63	0	99.3
<i>CD8A</i>	2	87 012 293	NM_001145873.1:c.*750T>C	rs3020729	99.15	0	99
<i>CD8A</i>	2	87 016 506	NM_001145873.1:c.565T>C	rs3020726	100	0	100
<i>TMEM127</i>	2	96 919 642	NM_017849.3:c.621G>A	rs3852673	47.73	0	100
<i>SNRNP200</i>	2	96 952 833	NM_014014.4:c.3550T>C	rs3171927	100	0	99.04
<i>AFF3</i>	2	100 210 074	NM_001025108.1:c.2124C>G	rs1047281	42.31	0	100
<i>AFF3</i>	2	100 210 642	NM_001025108.1:c.1556A>G	rs1047265	46.15	0	100
<i>NPAS2</i>	2	101 587 455	NM_002518.3:c.1059C>T	rs1562313	50	0	100
<i>NPAS2</i>	2	101 591 304	NM_002518.3:c.1180A>G	rs2305160	47.78	0	100

Table 3 (Continued)

Gene	Chromosome	Coordinate	HGVS	dbSNP ID	Father	Mother	Propositus
<i>NPAS2</i>	2	101 609 827	NM_002518.3:c.2130C>T	rs9223	45.22	100	0
<i>TGFBRAP1</i>	2	105 924 510	NM_001142621.1:c.249T>C	rs2241801	0	100	0
<i>RANBP2</i>	2	109 367 977	NM_006267.4:c.1456-7A>G	rs144029181	48.46	0	97.11
<i>RANBP2</i>	2	109 374 952	NM_006267.4:c.2550A>G	rs826549	46.55	0	100
<i>EDAR</i>	2	109 526 969	NM_022336.3:c.750C>T	rs260632	50	100	0
<i>BCL2L1</i>	2	111 907 691	NM_138621.4:c.465T>C	rs724710	0	100	0
<i>ANAPC1</i>	2	112 590 825	NM_022662.3:c.2448T>C	rs148461319	56.45	0	100
<i>DBI</i>	2	120 125 053	NM_001178017.1:c.141C>G	rs3731608	0	100	0
<i>DBI</i>	2	120 125 058	NM_001178017.1:c.146C>T	rs3731607	0	100	0
<i>BIN1</i>	2	127 816 632	NM_139343.2:c.957C>T	rs2276579	58.23	0	100
<i>BIN1</i>	2	127 821 499	NM_139343.2:c.698+10A>G	rs72481904	51.72	0	100
<i>BIN1</i>	2	127 826 533	NM_139343.2:c.486T>C	rs1060743	60.53	0	100
<i>MYO7B</i>	2	128 366 272	NM_001080527.1:c.2644-11G>A	rs17261971	44.44	0	98.94
<i>MYO7B</i>	2	128 379 563	NM_001080527.1:c.3454C>G	rs61744148	44	0	99.07
<i>MYO7B</i>	2	128 381 808	NM_001080527.1:c.3882C>T	rs13422424	48.41	0	100
<i>MYO7B</i>	2	128 392 688	NM_001080527.1:c.5636-7C>T	rs13391954	46.85	0	98.52
<i>RAB3GAP1</i>	2	135 893 372	NM_001172435.1:c.1793A>G	rs10445686	45.61	0	100
<i>LCT</i>	2	136 561 557	NM_002299.2:c.4606C>T	rs2304371	31.71	100	0
<i>LCT</i>	2	136 590 746	NM_002299.2:c.655G>A	rs3754689	46.67	0	100
<i>MMADHC</i>	2	150 432 976	NM_015702.2:c.453G>A	rs11545261	36.99	100	0
<i>NEB</i>	2	152 352 843	NM_001164507.1:c.24433G>C	rs7575451	100	0	100
<i>NEB</i>	2	152 387 553	NM_001164507.1:c.21585A>G	rs4664475	100	0	100
<i>NEB</i>	2	152 388 415	NM_001164507.1:c.21418-6delA	rs56026241	98.85	0	95.15
<i>NEB</i>	2	152 422 076	NM_001164507.1:c.18305G>C	rs2288210	99.27	0	100
<i>NEB</i>	2	152 422 087	NM_001164507.1:c.18294T>C	rs2288211	52.55	0	98.46
<i>NEB</i>	2	152 490 219	NM_001164507.1:c.9363T>C	rs6709886	100	0	97.09
<i>NEB</i>	2	152 527 608	NM_001164507.1:c.4435G>A	rs34577613	48.31	0	100
<i>ACVR1C</i>	2	158 443 889	NM_145259.2:c.114C>T	rs4556933	44.65	100	0
<i>SCN3A</i>	2	165 987 772	NM_006922.3:c.2547A>C	rs62174900	55.56	0	96.15
<i>SCN3A</i>	2	166 003 479	NM_006922.3:c.1441C>T	rs16850131	50.31	0	100
<i>SCN9A</i>	2	167 144 974	NM_002977.3:c.1287T>A	rs6747673	0	100	0
<i>TLK1</i>	2	171 906 469	NM_012290.4:c.844-10delT	rs3217296	0	100	0
<i>TLK1</i>	2	171 910 313	NM_012290.4:c.690G>A	rs11553951	0	100	0
<i>DCAF17</i>	2	172 338 113	NM_025000.3:c.*489T>G		55.85	0	100
<i>CYBRD1</i>	2	172 398 091	NM_024843.3:c.194-4C>G	rs62181678	52.38	0	100
<i>CYBRD1</i>	2	172 411 273	NM_024843.3:c.797G>A	rs10455	54.93	100	0
<i>SLC25A12</i>	2	172 693 780	NM_003705.4:c.466-3C>T	rs3765166	100	0	100
<i>ITGA6</i>	2	173 352 103	NM_001079818.1:c.2082C>T	rs2293649	0	100	0
<i>HOXD4</i>	2	177 016 728	NM_014621.2:c.367T>C	rs34727427	52.94	0	99.25
<i>TTN-AS1, TTN</i>	2	179 432 185	NM_001267550.1:c.78674T>C	rs12463674	34.18	0	100
<i>TTN-AS1, TTN</i>	2	179 440 029	NM_001267550.1:c.70830C>T	rs12464787	52.69	0	98.61
<i>TTN-AS1, TTN</i>	2	179 462 494	NM_001267550.1:c.57315T>C	rs35833641	45.69	0	99.8
<i>TTN</i>	2	179 497 133	NM_001267550.1:c.43488G>A	rs56034831	50.68	0	98.95
<i>TTN</i>	2	179 545 859	NM_001267550.1:c.33287G>A	rs36051007	63.89	0	100
<i>TTN</i>	2	179 650 701	NM_001267550.1:c.2244G>A	rs6715406	100	0	100
<i>ZNF385B</i>	2	180 308 028	NM_152520.4:c.1365G>A	rs2367796	0	100	0
<i>ITGA4, CERKL</i>	2	182 402 213	NM_000885.4:c.*1960_*1962delTTC		40.94	0	97.22
<i>CERKL</i>	2	182 521 578	NM_001030311.2:c.156C>T	rs1473295	0	100	0
<i>FRZB</i>	2	183 702 670	NM_001463.3:c.861+6C>A	rs111740973	50.82	0	100
<i>NCKAP1</i>	2	183 817 957	NM_205842.1:c.2274A>T	rs66829551	37.23	0	100
<i>COL3A1</i>	2	189 863 055	NM_000090.3:c.1977+10G>A		49.32	0	100
<i>STAT1</i>	2	191 874 667	NM_007315.3:c.63T>C	rs2066802	49.21	0	100
<i>CASP10</i>	2	202 082 459	NM_032977.3:c.1564T>A	rs13006529	0	100	0
<i>ALS2</i>	2	202 593 280	NM_020919.3:c.2796C>T	rs3219161	46.52	0	99.69
<i>ALS2</i>	2	202 598 113	NM_020919.3:c.2466G>A	rs2276615	51.02	100	0
<i>ICOS</i>	2	204 821 494	NM_012092.3:c.501+6C>T	rs4264550	27.78	0	100
<i>ICOS</i>	2	204 824 324	NM_012092.3:c.*2A>C	rs10183087	46.03	0	100
<i>PAR3B</i>	2	205 990 353	NM_152526.5:c.1326C>T	rs2276676	57.47	0	100
<i>FASTKD2</i>	2	207 652 763	NM_001136193.1:c.1697C>T		50	0	100
<i>MAP2</i>	2	210 518 057	NM_002374.3:c.163G>A	rs139310749	51.89	0	100
<i>UNC80</i>	2	210 705 348	NM_032504.1:c.3339C>G	rs10178675	54.79	0	100
<i>UNC80</i>	2	210 843 383	NM_032504.1:c.8871G>A	rs2075117	99.23	0	99.22

Table 3 (Continued)

Gene	Chromosome	Coordinate	HGVS	dbSNP ID	Father	Mother	Propositus
<i>ACADL</i>	2	211 060 050	NM_001608.3:c.997A>C	rs2286963	67.27	0	100
<i>ACADL</i>	2	211 074 909	NM_001608.3:c.603+11A>G	rs3764913	39.39	0	100
<i>CPS1</i>	2	211 456 637	NM_001122633.2:c.1048A>G	rs1047883	46.15	99.19	0
<i>CPS1</i>	2	211 456 639	NM_001122633.2:c.1050C>T	rs2229589	45.67	99.2	0
<i>CPS1</i>	2	211 481 257	NM_001122633.2:c.2697C>G	rs2287599	36.17	100	0
<i>SPAG16</i>	2	214 794 743	NM_024532.4:c.1274A>C	rs12623569	0	97.73	0
<i>BARD1</i>	2	215 595 203	NM_000465.2:c.1933T>C	rs2228456, rs34744268	30.95	0	98.47
<i>BARD1</i>	2	215 674 224	NM_000465.2:c.70C>T	rs1048108	100	0	100
<i>ABCA12</i>	2	215 852 421	NM_173076.2:c.3926A>G	rs145517253	53.33	0	98.44
<i>FN1</i>	2	216 300 482	NM_212482.1:c.44A>T	rs1250259	100	0	100
<i>PECR</i>	2	216 908 679	NM_018441.5:c.774G>A	rs2303842	46.91	0	99.07
<i>CXCR1</i>	2	219 029 108	NM_000634.2:c.827G>C	rs2234671	51.52	0	100
<i>SLC11A1</i>	2	219 252 630	NM_000578.3:c.747C>T	rs17221959	100	0	99.22
<i>SLC11A1</i>	2	219 259 732	NM_000578.3:c.1627G>A	rs17235409	42.55	0	100
<i>CYP27A1</i>	2	219 677 107	NM_000784.3:c.609G>A	rs148132118	32.91	0	100
<i>PRKAG3</i>	2	219 692 597	NM_017431.2:c.779A>G	rs41272689	54.44	0	100
<i>DES</i>	2	220 285 309	NM_001927.3:c.828C>T	rs1058261	44.83	98.51	0
<i>DES</i>	2	220 285 666	NM_001927.3:c.1014G>C	rs12920	48.65	96.3	0
<i>DES</i>	2	220 286 142	NM_001927.3:c.1104G>A	rs1058284	47.73	100	0
<i>OBSL1</i>	2	220 431 631	NM_015311.2:c.2055C>A	rs1043537	58.33	0	100
<i>OBSL1</i>	2	220 432 014	NM_015311.2:c.1818C>T	rs61732787	42.86	0	100
<i>PAX3</i>	2	223 161 889	NM_181459.3:c.129T>C	rs12623857	48.98	100	0
<i>CUL3</i>	2	225 434 445	NM_001257198.1:c.45T>C	rs10498163	42.5	0	97.73
<i>SP110</i>	2	231 036 866	NM_080424.2:c.1731C>T	rs13018234	32.05	0	100
<i>DIS3L2</i>	2	233 201 328	NM_152383.4:c.2646A>G	rs3811578	47.13	0	100
<i>UGT1A8, UGT1A10, UGT1A9, UGT1A7, UGT1A6, UGT1A5, UGT1A4, UGT1A3</i>	2	234 638 580	NM_019078.1:c.867+16076A>G	rs45449995	50	0	100
<i>COL6A3</i>	2	238 243 292	NM_004369.3:c.9206C>T	rs1131296	47.14	0	100
<i>MLPH</i>	2	238 427 194	NM_024101.5:c.458T>C	rs3751109	47.71	0	99.17
<i>MLPH</i>	2	238 427 251	NM_024101.5:c.515G>A	rs3751107	45.16	0	98.95
<i>HDAC4</i>	2	240 036 959	NM_006037.3:c.1566G>A	rs79199572	50	0	99.04
<i>CAPN10</i>	2	241 531 479	NM_023083.3:c.600A>G	rs3792269	0	100	0
<i>CAPN10</i>	2	241 536 126	NM_023083.3:c.1510A>G	rs7607759	0	100	0
<i>CAPN10</i>	2	241 538 061	NM_023083.3:c.1990-7T>C	rs55878652	0	100	0
<i>KIF1A</i>	2	241 680 802	NM_001244008.1:c.3641-8C>T	rs56024577	66.67	0	100
<i>KIF1A</i>	2	241 685 283	NM_001244008.1:c.3246C>T		42.86	0	97.78

HGVS, Human Genome Variation Society; SNP, single-nucleotide polymorphism.

data also support the testing of hepatic and non-hepatic VKD proteins in VKCFD1 cases to evaluate the effectiveness of VK₁ treatment of this disorder with bleeding and non-bleeding clinical phenotypes.

Finally, the non-severe bleeding phenotype in a patient with low levels of VKD hemostatic proteins, an INR of > 8 and an absence of response to VK₁ is intriguing. As neither the levels of VKD proteins nor the *GGCX* mutation seem to explain the bleeding phenotype [1,9], variability in other hemostatic factors, particularly elements involved in primary hemostasis or fibrinolysis, might modulate the bleeding phenotype in VKCFD patients. We speculated that the same mechanism causing the disease, the UPD, could also have contributed to the non-severe bleeding. Any functional genetic variant of chromosome 2 inherited from the father will be in homozygosis in the proband. An

interesting candidate is *TFPI* [19]. However, no mutation was identified in the *TFPI* exons. Alternatively, mutations in other genes involved in related diseases, such as pseudoxanthoma elasticum (*ABCC6*), Keutel syndrome (*MGP*), or VKCFD2 (*VKORC1*), might also modulate the clinical phenotype of patients with VKCFD1 [1,8,9]. Further studies are necessary to identify the mechanism(s) modulating the clinical effects of mutations causing VKCFD1.

Addendum

M. A. Dasi, S. Izquierdo, and B. Argilés attended to the patient and relatives, supervised the follow-up, and were responsible for the clinical and coagulation data. J. Padilla and N. Garcia-Barberá were responsible for

sequencing and RT-PCR. J. L. Garcia, J. M. Hernández-Sánchez, and J. M. Hernández-Rivas performed the CGH and NGS assays, and analyzed the results. M. E. de la Morena-Barrio was responsible for the *in silico* study. J. Corral and R. Gonzalez-Conejero designed the research plan and coordinated the experimental work. V. Vicente contributed to critical revision. V. Vicente, J. Corral, and R. Gonzalez-Conejero interpreted the results and wrote the manuscript.

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Disclosure of Conflict of Interests

The authors state that they have no conflict of interest.

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